Vehicle Routing & Delivery Planning

Synthetic Dataset - Full Overview & Module Guide

VRPTW workflow | Seed 20261022 | 500 sites | 2,000 deliveries | 100 vehicles | 30 days

1. Ye project kya hai?

Ye ek **synthetic data generator + validation framework** hai jo ek delivery-planning (Vehicle Routing Problem with Time Windows, VRPTW) workflow ka poora dataset banata hai.

- **Input:** ek seed (20261022) aur scale parameters (500 sites, 2,000 deliveries, 100 vehicles, 30 days)
- **Process:** master data, transactions, routes, simulated outcomes, anomaly injection, validation
- **Output:** CSV/JSON/SQL files, ground truth labels, data dictionary, README aur validation report

Iska goal ek aisa dataset dena hai jisse analyst ya ML model **route quality, constraint violations aur delivery failures** evaluate kar sake.

2. End-to-end flow

```
Config (seed, scale)
   |
   v
Master Data: CustomerSite, Vehicle, VehicleCapacity, Driver, Shift, TravelTimeMatrix
   |
   v
Demand: Order -> Delivery -> TimeWindow
   |
   v
Planning: Route -> RouteStop (capacity, shift, window rules apply)
   |
   v
Simulation: actual arrivals -> DeliveryOutcome
   |
   v
Anomaly Injection (labelled) -> ground_truth.csv
   |
   v
Validation -> Export -> Reproducibility check (2 reruns)
```

3. Entity-wise (module-wise) explanation

A. Master Data (pehle banta hai)

Entity	Kaam	Key fields / rules
CustomerSite	Delivery location (500 sites)	site_id, lat/lon, region, address type, access restrictions (e.g. truck size limit). Coordinates ek defined region ke andar hote hain.
Vehicle	Fleet ke 100 vehicles	VEH-000001, type (van/truck/mini), status, restricted zones, effective dates.

Entity	Kaam	Key fields / rules
VehicleCapacity	Har vehicle ki capacity limits	weight (kg), volume (m3), max stops. Alag table kyunki capacity time ke saath badal sakti hai (effective dates).
Driver	Driver master	DRV-000001, license class, home depot, active status.
Shift	Driver ka duty window	shift start/end, break duration, max driving hours. Driver 1:N Shift.
TravelTimeMatrix	Site-to-site distance aur time	30 din ke traffic scenarios (peak / off-peak / disruption). Distance haversine se; time = distance / speed x traffic factor.

B. Demand Modules

Entity	Kaam	Key fields / rules
Order	Customer ki demand	ORD-2026-000001, site FK, quantity, weight, volume, priority, lifecycle (created, planned, completed/cancelled).
Delivery	Order ka executable delivery task (2,000)	DEL-2026-000001, order FK, service duration, required vehicle type. Ek Order me multiple Deliveries ho sakti hain (1:N).
TimeWindow	Delivery ka allowed arrival slot	earliest/latest time, hard/soft type, priority. Boundary cases bhi include hote hain.

C. Planning Modules

Entity	Kaam	Key fields / rules
Route	Ek vehicle + driver + shift ka din bhar ka plan	RTE-2026-000001, vehicle/driver/shift FK, total load, total distance, planned duration. Totals child stops se **calculate** hote hain.
RouteStop	Route ke andar ordered stops	STOP-<route-id>-001, sequence number, planned arrival/departure, delivery FK. Route 1:N RouteStop.

Delivery aur Route ka relation effectively **N:M** ban sakta hai (re-planning ke time), isliye RouteStop bridge ka kaam karta hai aur history preserve hoti hai. Jo deliveries assign nahi hoti, unko explicitly **unserved + reason** mark kiya jata hai.

D. Simulation / Outcome Module

Entity	Kaam	Key fields / rules
DeliveryOutcome	Actual result	actual arrival/departure, status (delivered / late / failed / skipped), reason code, delay minutes.

Outcomes **rules se nikalte hain, random nahi**. Agar actual arrival time window se bahar hai to *late* status banta hai. Agar customer absent ho ya site access restricted ho to *failed* status aur reason code milta hai.

4. Business rules jo enforce hote hain

- Route ka total load <= vehicle capacity
- Stop arrival >= previous stop departure + travel time
- Arrival time window ke andar (ya late flag)
- Driver shift ke andar ya shift-exceeded flag
- Restricted vehicle restricted site par nahi jayega
- Har delivery ya assigned hai ya explicitly unserved

- Failed delivery ke paas reason code zaroori hai
- State changes naye event records me store hote hain, purane overwrite nahi hote

5. Anomaly Injection Module (negative scenarios)

Pehle **clean baseline** banta hai, phir chhote percentage me labelled errors inject hote hain:

Anomaly	Kya hota hai
Route exceeds capacity	Load vehicle limit se zyada
Stop omitted / duplicated	Delivery route me missing ya do baar
Arrival before previous departure	Time sequence galat
Driver shift exceeded	Route shift end ke baad tak chalta hai
Coordinate outside region	Site lat/lon defined boundary ke bahar

Har anomaly ke liye **issue_type**, affected ID aur expected decision **sirf ground_truth.csv** me jaata hai. Input tables me koi leakage nahi hota, aur unrelated rows corrupt nahi hote.

6. Validation Module

Ye automated checks chalata hai:

1. Primary-key uniqueness
2. Foreign-key integrity (saare joins valid)
3. Enum values allowed set me
4. Date/time ordering
5. Arithmetic reconciliation (route totals = stop totals)
6. Lifecycle transitions valid
7. Duplicate detection
8. Injected anomalies ko detect karke ground truth se match karna

Result **validation_summary** me pass/fail counts ke saath aata hai.

7. Metrics (measurable outputs)

- On-time delivery rate
- Failed delivery rate (reason code ke hisaab se)
- Capacity utilization per vehicle
- Total distance aur total travel time
- Average delay minutes
- Unserved delivery count
- Shift utilization aur overtime count
- Anomaly detection precision / recall

8. Reproducibility Module

- Fixed seed **20261022**; scale parameters CLI se control hote hain (starter vs full dataset, same generator)

- Generator command ko 2 baar run karte hain
- Row counts aur file checksums (SHA-256) compare hote hain, aur match hona chahiye

9. Final deliverables

```
/data         -> CSVs (12 entities) + JSON + SQL seed
/ground_truth -> ground_truth.csv
/docs         -> README, data_dictionary, schema (ERD), source_manifest
/validation   -> validation_summary
/generator    -> generate.py + command + seed config
```

Source manifest me likha hota hai ki **synthetic data hai, koi external/personal data use nahi hua** (lawful provenance).

10. Actors ko kya milta hai

Actor	Output / benefit
Planner	Routes aur unserved deliveries ki list
Warehouse operator	Load aur stop sequence
Logistics analyst	Metrics aur anomaly analysis
Operations manager	KPIs (on-time %, failure %, utilization)
Supplier	Order-to-delivery traceability